

# ■ CRT LOCK — 2-Speaker Traffic Signal Presentation Script

Format: 2-Speaker Dialogue / Conversation

Characters:

- Alex (Speaker 1): The Everyday Driver / Host (Curious, relatable, asks questions)
- Sam (Speaker 2): The Traffic Tech Lead (Explains concepts simply, runs the live demo)

Target Duration: ~3–4 Minutes

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## ■ Scene 1: The Problem (Stuck in Traffic)

Alex:

> *"Sam, be honest with me... why does it feel like whenever I'm in a rush, every single traffic light turns red right in front of me?"*

Sam:

> *"Haha! It's actually a classic math problem. Traffic signals along a busy street operate on different timer cycles — like 3 minutes, 5 minutes, or 7 minutes. If they aren't synchronized, you get stuck stopping at every block."*

Alex:

> *"So how do city planners fix that? How do you get a 'Green Wave' where a car can drive straight through all green lights?"*

Sam:

> *"That's where a 2,000-year-old math theorem called the Chinese Remainder Theorem comes in. And to prove it, we built this interactive visual app called CRT Lock!"*

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## ■ Scene 2: The Traffic Signal Analogy

Sam: *\*(Pointing to the app screen at `http://localhost:5173/`)*

> *"Imagine our street has 3 traffic signals:"*

> - *\*Signal 1 changes every 3 minutes.\**

> - *\*Signal 2 changes every 5 minutes.\**

> - *\*Signal 3 changes every 7 minutes.\**

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> *"In our app, each rotating gear is a traffic signal. The slot in the gear represents a GREEN LIGHT, and this sliding key rod is your car trying to drive through."*

Alex:

> *"Got it! So if all three gear slots line up, the rod slides through — meaning my car hits green lights across the whole street?"*

Sam:

> *"Spot on! But if even ONE signal is red, the gear blocks the rod — CLACK! — and your car is forced to stop."*

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## ■ Scene 3: Live Interactive Demo

Alex:

> \*"Alright, let's test it! Suppose Signal A turned green 2 minutes ago, Signal B turned green 3 minutes ago, and Signal C turned green 2 minutes ago. What minute should I dial?"\*

Sam:

> \*"Let's try a random guess first. Give me a number on the dial."\*

Alex:

> \*"Let's try Minute 20!"\*

Sam: \*(Sets main dial to 20)\*

> \*"Okay! At Minute 20:"\*

> - \*Signal 1 (3-min timer): GREEN light!\*

> - \*Signal 2 (5-min timer): RED light!\*

> - \*Signal 3 (7-min timer): RED light!\*

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> \*Let's hit Check Lock to try driving through..."\*

> \*(Clicks 'Check Lock' — Loud THUD sound as rod hits a dead stop)\*

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> \*"BAM! Red light at Intersection #2! Your car is stuck."\*

Alex:

> \*"Ouch! Okay, so Minute 20 doesn't work. How do we find the exact minute when ALL THREE turn green?"\*

Sam:

> \*"That's what the Chinese Remainder Theorem calculates for us. For our 3, 5, and 7 minute timers, the exact magic moment is Minute 23! Dial it to 23."\*

Alex: \*(Dials to 23)\*

> \*"Look! All three signal lights on the dashboard turned GREEN!"\*

Sam:

> \*"Now let's drive through..."\*

> \*(Clicks 'Check Lock' — Smooth sliding sound, spring latch pops open!)\*

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> \*"The rod glides straight through all three gears! Click Open..."\*

> \*(Clicks 'Open' — Door swings open with victory chime)\*

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> \*"The door opens revealing 23 — the exact green wave minute!"\*

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## ■ Scene 4: Wrap-Up & Everyday Impact

Alex:

> \*"That's awesome! So the Chinese Remainder Theorem basically finds the one secret timing where different repeating cycles align?"\*

Sam:

> \*"Exactly. And beyond traffic lights, engineers use this exact same math for fiber-optic internet cables, computer chip processing, and cybersecurity encryption."\*

Alex:

> \*"It turns heavy textbook math into a visual puzzle you can actually see and hear. Thanks everyone!"\*

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■ Speaker Role Assignments Summary

| Role | Speaker Name | Focus |

| :--- | :--- | :--- |

| Speaker 1 | Alex | Asks everyday questions, reacts to the demo, plays the relatable driver. |

| Speaker 2 | Sam | Operates the CRT Lock app, explains the math/gears, drives the solution. |